Data Center Lifetime Digital Twin

Connect AI scheduling to equipment life, power capacity and operating cost.

01 / OBSERVE

Your operating data center

AI workloads + GPU racks



Facility power assets



Jobs, power, temperature + health

Joined by time and asset identity

Measured load + state

02 / PREDICT

One twin. Four asset-life models.

Transformers

Hot-spot temperature

Paper aging + oil condition

Separate thermal and chemical models

Gas turbines

Load + thermal cycling

Fatigue + creep assessment

Component-specific calibration

BESS

SOC + cell temperature

Capacity fade + resistance

Calendar aging + cycle aging

Power electronics

Junction temperature

Thermal-cycle damage

Device and package-specific curves

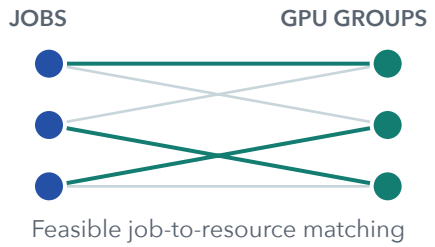
Shared power network, cooling and retained asset states

Forecasts + uncertainty

Test plans in the twin

03 / OPTIMIZE

Coordinate the decisions



Feasible job-to-resource matching

- ✓ Place or stagger AI jobs
- ✓ Dispatch turbines + BESS
- ✓ Meet deadlines + reserves

Safety checks + operator approval

04 / APPROVED ACTIONS THROUGH EXISTING SCHEDULER + ENERGY MANAGEMENT

POTENTIAL IMPACT / TO BE VALIDATED WITH SITE DATA

Plan capacity confidently

Test GPU growth against power and thermal headroom.

Reduce lifecycle cost

Balance energy, equipment wear and maintenance tradeoffs.

Strengthen resilience

Stress-test ramps, outages and backup-power reserves.

Protect compute value

Meet service targets while shaping infrastructure demand.

Open-source foundation: ExaDigiT/RAPS | pandapower + ANDES | PyBaMM / BLAST-Lite | OpenModelica | pyLife | OR-Tools / HiGHS

Concept architecture. Benefits require calibration and validation. Existing equipment controls and protection remain authoritative.